

GREAT version 4.0.4 current (08/19/2019 to now)

▼

Warning: Your set hits a large fraction of the genes in the genome, which often does not work well with the GREAT Significant by Both view due to a saturation of the gene-based hypergeor
See our [tips for handling large datasets](#) or try the [Significant By Region-based Binomial](#) view.

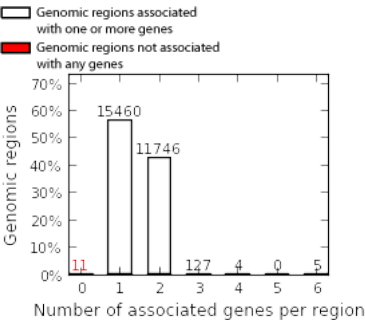
Job Description

Region-Gene Association Graphs

What do these graphs illustrate?

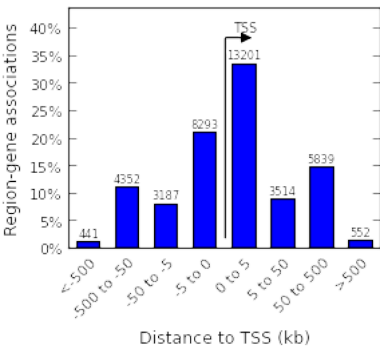
Number of associated genes per region

Download as PDF.



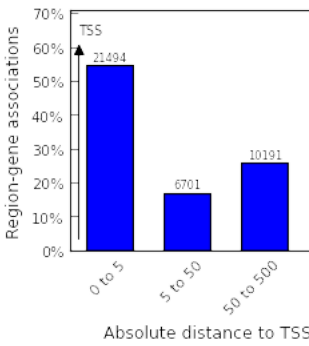
Binned by orientation and distance to TSS

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Binned by absolute distance to TSS

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Global Controls Global Export ▼

Which data is exported by each option?

Ensembl Genes (no terms)

GO Biological Process (20+ terms)

Table controls: Export ▼ Shown top rows in this table: 20 Set Term annotation count: Min: 1 Max: Inf Set Visualize this table: [select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
endoplasmic reticulum unfolded protein response	31	6.6580e-37	2.8262e-34	2.0787	373	1.36%	544	1.9449e-5	1.3627	92
snRNA metabolic process	55	2.1239e-28	5.0815e-26	2.2047	247	0.90%	748	4.8814e-4	1.3548	69
regulation of pri-miRNA transcription from RNA polymerase II promoter	62	1.2927e-27	2.7436e-25	2.2335	233	0.85%	804	8.7237e-4	1.5721	27
DNA damage response, signal transduction by p53 class mediator resulting in cell cycle arrest	73	1.6903e-25	3.0469e-23	2.1007	246	0.90%	507	8.5779e-6	1.4979	55
signal transduction involved in mitotic G1 DNA damage checkpoint	77	3.6696e-25	6.2713e-23	2.0812	248	0.91%	486	5.9308e-6	1.5010	56
ncRNA transcription	81	4.4879e-25	7.2910e-23	2.0022	272	0.99%	515	9.4260e-6	1.4133	77
cellular component disassembly involved in execution phase of apoptosis	84	2.8935e-24	4.5328e-22	3.0957	113	0.41%	1,172	1.3850e-2	1.4937	23
snRNA transcription from RNA polymerase II promoter	86	4.0126e-24	6.1397e-22	2.1355	223	0.82%	724	3.5284e-4	1.3991	58
apoptotic nuclear changes	110	1.0267e-21	1.2283e-19	3.0015	105	0.38%	1,314	2.7927e-2	1.4775	21
positive regulation of pri-miRNA transcription from RNA polymerase II promoter	115	5.2109e-21	5.9627e-19	2.5311	135	0.49%	1,017	5.3526e-3	1.6041	19
cytoplasmic translation	117	8.5383e-21	9.6030e-19	2.6128	126	0.46%	446	2.1713e-6	1.6100	41
IRE1-mediated unfolded protein response	121	1.2817e-20	1.3939e-18	2.2104	174	0.64%	756	5.3521e-4	1.4266	49
regulation of receptor biosynthetic process	130	6.2865e-20	6.3634e-18	3.0792	92	0.34%	1,331	2.9073e-2	1.5197	18
cytoplasmic sequestering of protein	144	3.9118e-19	3.5747e-17	2.6392	113	0.41%	1,260	2.2004e-2	1.3855	32
positive regulation of nuclear-transcribed mRNA catabolic process, deadenylation-dependent decay	193	1.5988e-16	1.0901e-14	2.3285	122	0.45%	1,307	2.7608e-2	1.5830	15
ER-nucleus signaling pathway	225	3.7956e-15	2.2199e-13	2.1010	139	0.51%	1,077	8.4927e-3	1.4474	30

Name	Rank	P-Value	FDR Q-Val	Enrichment	Region Hits	Coverage	Rank	FDR Q-Val	Enrichment	Gene Hits
positive regulation of nuclear-transcribed mRNA poly(A) tail shortening	278	3.0961e-13	1.4655e-11	2.5975	77	0.28%	1,342	3.0751e-2	1.6886	11
ER overload response	280	3.3006e-13	1.5512e-11	3.1596	56	0.20%	1,342	3.0751e-2	1.6886	11
hepatocyte apoptotic process	322	4.1695e-12	1.7039e-10	2.5477	72	0.26%	1,229	1.9878e-2	1.6886	12
histone H2A ubiquitination	359	2.2658e-11	8.3051e-10	2.0934	101	0.37%	1,174	1.3839e-2	1.5351	20

The test set of 27,353 genomic regions picked 11,120 (59%) of all 18,777 genes.
GO Biological Process has 13,159 terms covering 16,804 (89%) of all 18,777 genes, and 1,256,055 term - gene associations.
13,159 ontology terms (100%) were tested using an annotation count range of [1, Inf].

GO Cellular Component (7 terms)

Table controls:

Export

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Term annotation count: Min: 1Max: InfSet

Visualize this table: [select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
COPI-coated vesicle	78	3.6570e-12	8.1016e-11	2.3587	84	0.31%	200	6.8720e-3	1.5417	21
nuclear ubiquitin ligase complex	79	7.8905e-12	1.7259e-10	2.0832	107	0.39%	257	3.0629e-2	1.3509	32
mitochondrial large ribosomal subunit	82	1.8730e-11	3.9470e-10	2.0593	106	0.39%	280	4.6402e-2	1.2913	39
intrinsic component of mitochondrial outer membrane	94	1.1241e-10	2.0665e-9	2.2999	76	0.28%	249	2.8891e-2	1.4683	20
integral component of mitochondrial outer membrane	97	2.3333e-10	4.1567e-9	2.2902	74	0.27%	272	3.9276e-2	1.4583	19
COPI-coated vesicle membrane	152	3.7887e-7	4.3071e-6	2.0709	59	0.22%	234	2.0250e-2	1.5830	15
CD40 receptor complex	197	8.9994e-6	7.8939e-5	2.4143	32	0.12%	240	2.2580e-2	1.6886	11

The test set of 27,353 genomic regions picked 11,120 (59%) of all 18,777 genes.
GO Cellular Component has 1,728 terms covering 17,911 (95%) of all 18,777 genes, and 382,522 term - gene associations.
1,728 ontology terms (100%) were tested using an annotation count range of [1, Inf].

GO Molecular Function (5 terms)

Table controls:

Export

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Term annotation count: Min: 1Max: InfSet

Visualize this table: [select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
mitogen-activated protein kinase binding	17	3.4213e-17	8.4970e-15	2.3069	130	0.48%	124	4.7474e-4	1.6260	26
myosin V binding	20	2.0763e-15	4.3830e-13	2.6896	86	0.31%	215	3.3700e-2	1.5893	16
NF-kappaB binding	26	5.8021e-14	9.4217e-12	2.2458	109	0.40%	115	3.1352e-4	1.6283	27
cyclin-dependent protein serine/threonine kinase inhibitor activity	49	4.9884e-11	4.2982e-9	2.9663	50	0.18%	220	3.5628e-2	1.6886	12
protein kinase B binding	57	2.4401e-10	1.8074e-8	3.1178	43	0.16%	220	3.5628e-2	1.6886	12

The test set of 27,353 genomic regions picked 11,120 (59%) of all 18,777 genes.
GO Molecular Function has 4,222 terms covering 16,729 (89%) of all 18,777 genes, and 230,772 term - gene associations.
4,222 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Human Phenotype (1 term)

Table controls:

Export

Shown top rows in this table: 20Set

Term annotation count: Min: 1Max: InfSet

Visualize this table: [select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
Death in infancy	1	2.4361e-22	1.6256e-18	2.1741	197	0.72%	296	3.7123e-4	1.4474	48

The test set of 27,353 genomic regions picked 11,120 (59%) of all 18,777 genes.
Human Phenotype has 6,673 terms covering 3,390 (18%) of all 18,777 genes, and 255,152 term - gene associations.
6,673 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Mouse Phenotype Single KO (9 terms)

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
abnormal bone mineral density of femur	24	1.9006e-13	7.2516e-11	2.8752	66	0.24%	965	5.0268e-2	1.6886	10
decreased cellular sensitivity to ultraviolet irradiation	42	4.7881e-12	1.0439e-9	2.9800	55	0.20%	782	2.1739e-2	1.6886	12
delayed embryo turning	44	5.3603e-12	1.1156e-9	2.1406	102	0.37%	728	1.5530e-2	1.5351	20
increased incidence of tumors by ionizing radiation induction	78	4.9482e-10	5.8090e-8	2.5921	56	0.20%	929	4.3045e-2	1.5760	14
embryonic lethality between somite formation and embryo turning, incomplete penetrance	106	7.0970e-9	6.1309e-7	2.1588	70	0.26%	695	1.4109e-2	1.5948	17
abnormal hypoglossal nerve topology	129	2.9541e-8	2.0970e-6	2.0668	71	0.26%	695	1.4109e-2	1.5948	17
absent hypoglossal nerve	168	1.3273e-7	7.2347e-6	2.2264	54	0.20%	929	4.3045e-2	1.5760	14
thin hypoglossal nerve	200	2.6678e-7	1.2214e-5	2.0517	62	0.23%	571	6.1748e-3	1.6886	15
reduced sympathetic cervical ganglion size	387	2.2607e-5	5.3492e-4	2.0027	44	0.16%	703	1.4315e-2	1.6886	13

The test set of 27,353 genomic regions picked 11,120 (59%) of all 18,777 genes.
Mouse Phenotype Single KO has 9,157 terms covering 9,525 (51%) of all 18,777 genes, and 563,371 term - gene associations.
9,157 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Mouse Phenotype (11 terms)

Table controls:

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 Term annotation count: Min:

1

 Max:

Inf

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 Visualize this table:

[select one]

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits
increased hemangioma incidence	20	5.4273e-19	2.5945e-16	2.3231	143	0.52%	617	3.2891e-4	1.5796	29
decreased cellular sensitivity to ultraviolet irradiation	51	9.9525e-15	1.8658e-12	2.9300	71	0.26%	856	4.3007e-3	1.6886	15
increased renal carcinoma incidence	75	1.7714e-12	2.2582e-10	3.6610	43	0.16%	1,330	3.8082e-2	1.6886	10
abnormal frontal lobe morphology	87	5.5121e-12	6.0576e-10	2.3256	85	0.31%	1,275	3.2747e-2	1.5760	14
increased kidney tumor incidence	98	2.1843e-11	2.1310e-9	2.4898	70	0.26%	1,183	2.2162e-2	1.5830	15
abnormal small intestinal crypt cell proliferation	99	2.8419e-11	2.7446e-9	2.0761	102	0.37%	1,044	1.1307e-2	1.5351	20
abnormal hypoglossal nerve topology	225	2.9541e-8	1.2553e-6	2.0668	71	0.26%	1,019	1.0047e-2	1.5948	17
absent hypoglossal nerve	275	1.3273e-7	4.6147e-6	2.2264	54	0.20%	1,275	3.2747e-2	1.5760	14
thin hypoglossal nerve	308	2.6678e-7	8.2814e-6	2.0517	62	0.23%	856	4.3007e-3	1.6886	15
abnormal base-excision repair	346	8.1333e-7	2.2475e-5	2.4388	39	0.14%	1,330	3.8082e-2	1.6886	10
reduced sympathetic cervical ganglion size	553	2.2607e-5	3.9086e-4	2.0027	44	0.16%	1,024	1.0261e-2	1.6886	13

The test set of 27,353 genomic regions picked 11,120 (59%) of all 18,777 genes.
Mouse Phenotype has 9,561 terms covering 9,709 (52%) of all 18,777 genes, and 718,378 term - gene associations.
9,561 ontology terms (100%) were tested using an annotation count range of [1, Inf].



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Stanford Engineering

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